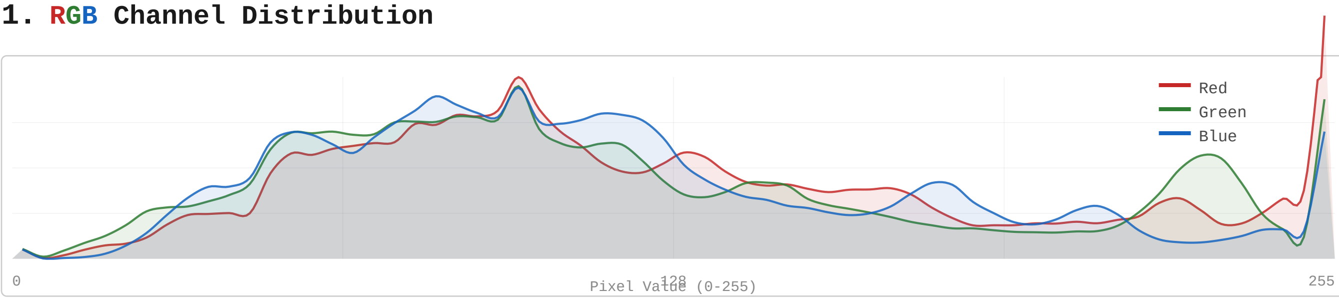
Decorrelation Gap: KODIM08

768x512 | 24-bit RGB | PNG (lossless)

Sunlit Trees and Pathway | Strongly one-dimensional (PC1 = 96.96%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

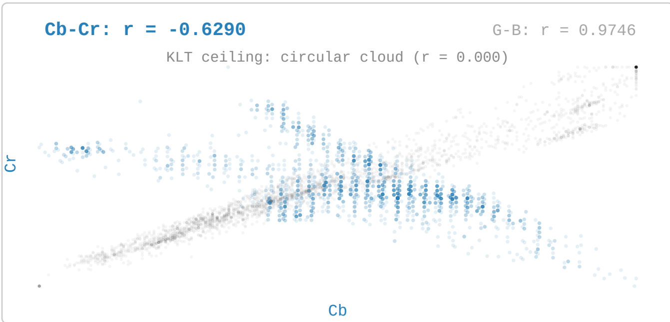
	R	G	B
R	1.0000	0.9665	0.9162
G	0.9665	1.0000	0.9746
B	0.9162	0.9746	1.0000

Cb-Cr:
-0.629
(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.4779	-0.0596
Cb	-0.4779	1.0000	-0.6290
Cr	-0.0596	-0.6290	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.966460	-0.477922	0.4779
R-B → Y-Cr	0.916208	-0.059590	0.0596
G-B → Cb-Cr	0.974616	-0.628992	0.6290
Avg |r|	0.952428	0.388835	0.3888

Decorrelation Gap: KODIM08

768x512 | 24-bit RGB | PNG (lossless)

Sunlit Trees and Pathway | Strongly one-dimensional (PC1 = 96.96%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

59.2% Moderate

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

4.7%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	42.34 dB
G	47.62 dB
B	40.77 dB
Overall	42.75 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5879	-0.6130	-0.5279	11265.86
PC2	-0.7265	0.1130	0.6778	310.20
PC3	0.3558	-0.7820	0.5118	43.16

8. PCA Derived Metrics

Condition Number

261.00

PC1/PC2: 36.32 | PC2/PC3: 7.19

Blue Channel Independence

4.7%

Variance not captured by PC1

PC1 Dominant Channel

G

Loading: G (0.61)

9. Redundancy Profile

Classification: Strongly one-dimensional | CN: 261.00 | PC1: 97.0%

RGB Correlation: Avg |r| = 0.952428 | R-G: 0.9665 | R-B: 0.9162 | G-B: 0.9746

YCbCr Residual: Avg |r| = 0.388835 | Y-Cb: -0.4779 | Y-Cr: -0.0596 | Cb-Cr: -0.6290

Efficiency: 59.2% of KLT ceiling | Gap: 0.3888 avg |r| remaining

Blue Channel: 4.7% independent | 4:2:0 PSNR: 42.75 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.